
SAMARTH : Web-Based Human-in-the-Loop Architecture for DBT Implementation under PCR & PoA Acts

OVERVIEW

SAMARTH is a secure, scalable, and transparent **web-based Direct Benefit Transfer system** designed for the effective implementation of the **PCR Act, 1955** and **PoA Act, 1989**. It integrates with existing **Digital Public Infrastructure (DPI)** — including **Aadhaar**, **DigiLocker**, **PFMS**, **eCourts**, and **CCTNS** — to automate data verification and workflow tracking, while ensuring that **final decisions remain human-controlled**.

The architecture is modular and layered to ensure **auditability, interoperability, and future scalability**.

LAYER 0 — User Interface & Access Layer

“Web portal deployed with load balancing and stateless design, supporting peak traffic and failover.”

Role:

Acts as the entry point for all stakeholders (officials, citizens, auditors). Handles UI rendering, authentication, access control, role-based dashboards, and relief selection interfaces.

(A) Government Officials Interface

Purpose:

Enable district, state, and central officers to review cases, verify documents, approve sanctions, track fund disbursement, and monitor multi-relief applications.

Key Modules:

- 1. Login & Role-Based Access Control (RBAC):**
 - Integrated with NIC eGov Single Sign-On (SSO) or State LDAP servers.
 - Session management via HTTPS + JWT; 2FA enforced for officials.
 - Roles: District Welfare Officer (DWO), District Magistrate (DM), Treasury Officer, Auditor, Administrator.
 - Access rights defined per role (view/edit/approve/fund release).
- 2. Dashboard View:**
 - Displays pending cases, deadlines, SLA status, and available funds.
 - Real-time counters categorized by Stage (Registered, Verified, Approved, Fund Released) and by relief type selected by users.

3. **Case Management Panel:**
 - Lists all incoming cases from CCTNS/eCourts (via Layer 1 APIs).
 - AI-assisted flagging for missing documents, duplicate FIRs, or inconsistencies. **(optional)**
 - Officers can verify, request clarification, or approve each relief type separately.
 4. **Document Viewer & Annotation Tool:**
 - Fetches caste certificate, ID, FIR, court orders, and other user-selected documents via DigiLocker/Aadhaar APIs.
 - Allows manual approval or rejection per relief type.
 - Audit logs every click and decision for full transparency.
 5. **Approval Workflow:**
 - Officers use digital eSign (eMudhra/eSign 2.0 API) to approve reliefs.
 - Approval triggers Layer 2 orchestration for stage-wise fund disbursement.
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(B) Citizen (Victim) Portal

Purpose:

Provide a transparent, user-friendly interface for citizens to select reliefs, upload documents, track status, and file grievances.

Key Modules:

1. **User Registration & eKYC:**
 - eKYC via Aadhaar Authentication API (UIDAI Auth 2.5).
 - OTP or biometric login for beneficiaries.
 - Optimized for India's 4G/5G; assisted access at district facilitation centers for remote users.
 2. **Document Upload via DigiLocker:**
 - Citizens select relevant documents for each relief type.
 - Selected files imported into Samarth's Document Intake Module.
 - Each DigiLocker login generates consent receipts stored in Layer 4 audit logs (timestamped and hashed).
 - Auto-verification and hash validation done in Layer 1.
 3. **Relief Selection & Status Tracker:**
 - Users choose applicable reliefs (Monetary, Housing, Education, etc.).
 - Real-time status per relief type: Submitted → Under Verification → Approved → Fund Released.
 4. **Grievance Module:**
 - Raise complaints or follow up on pending cases.
 - Integrated with Layer 4 Grievance Engine.
 5. **Multilingual & Accessibility Support:**
 - Available in English + 12 Indian languages.
 - Voice-assisted interface via Bhashini API.
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Security & Access Control

- HTTPS + JWT-based session management.
- 2FA for officials.
- OAuth 2.0 with NIC SSO for authentication.
- Logs user actions → sent to **Layer 4** (Audit).

LAYER 1 — Data Ingestion & Verification Layer

“Ingestion and validation modules are horizontally scalable, using distributed queues to handle large case volumes efficiently.”

Role:

Collects and validates all case-related data from DPI (CCTNS, eCourts, PFMS, Aadhaar, DigiLocker) while ensuring the correct documents are linked to the case. Humans (citizens/officers) guide the selection to avoid errors.

Core Data Sources & APIs

Source	Integration API	Function
CCTNS	State Police Crime Data API	Fetch FIR details, victim info, IPC/PoA sections
eCourts	eCourts REST API	Fetch judicial progress, case number, court orders
Aadhaar	UIDAI Authentication API	eKYC, demographic verification
DigiLocker	DigiLocker REST v2	Pull documents selected by user (caste certificate, FIR, court orders)
PFMS	Beneficiary Validation API	Validate bank account linkage and fund eligibility
District Portal	CSV/JSON Upload Interface	Manual intake for offline districts

Core Components

- Data Aggregator Engine:**
 - Pulls all case-related data from CCTNS, eCourts, Aadhaar, PFMS, and other state/central databases relevant to each relief type and **user-selected documents from DigiLocker**.
 - Creates one unified **Master Case Object**.
- Validation Engine:**
 - Checks data integrity, Aadhaar mapping, and document authenticity.
 - Ensures **only the documents selected by the user are fetched**.
 - Performs hash-based verification to ensure documents are unaltered.

Checks **service-specific criteria**:

3. Immediate Monetary Relief → FIR + bank account
 4. Relief & Rehabilitation Grant → FIR + medical/support documents
 5. Compensation & Disbursement → FIR + proof of loss
 6. Housing Assistance → FIR + property documents
 7. Self-Employment Subsidy → FIR + skill/training certificates
 8. Educational Assistance Grant → FIR + school/college enrollment proofs
 9. Inter-Caste Marriage Incentive → FIR + marriage certificate + caste certificates
 10. Funeral Assistance → FIR + death certificate
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 11. **Deduplication Module**:
 - Detects repeated FIRs, duplicate IDs, or multiple claims.
 - Ensures only one case per victim proceeds.
 12. **Queue Manager (Kafka / RabbitMQ)**:
 - Handles asynchronous ingestion and ensures no data is lost if Layer 1 fails.
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Workflow (Updated for DigiLocker User Selection)

1. **Citizen uploads or selects documents**:
 - On the portal, the victim **selects documents from DigiLocker** (via OAuth consent).
 - Each document gets a unique ID, hash, and metadata stored in the system.
2. **Case data pulled automatically**:
 - FIR from CCTNS, court orders from eCourts, Aadhaar verification, bank account validation.
 - Service-specific verification modules pull data from **Housing, Education, Employment, and Social Welfare databases**.
3. **Master Case Object created**:
 - Combines all verified information and **selected documents only**.
4. **Deduplication & verification**:
 - Checks for duplicates and validates document integrity (hashes, Aadhaar match).
5. **Case ready for human approval**:
 - Verified Master Case Object is passed to **Layer 2**.

Outcome:

- No AI guessing.
 - Only human-approved document selections proceed.
 - Reliable, auditable, and realistic for MVP/demo purposes.
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LAYER 2 — Workflow Orchestration & Human Approval Layer (Detailed Flow)

Workflow engine is cluster-ready, ensuring human approvals and fund release continue without downtime.”

Purpose:

This layer is the “control center” of AASHRAY.

It acts as a **coordinator between data verification (Layer 1) and financial disbursement (Layer 3)** while ensuring that **no automated fund release happens without human**

It supports **two categories of workflows**:

Category	Example Schemes	Trigger Type
Case-Linked (FIR-based)	Immediate Relief, Compensation, Rehabilitation	FIR + CCTNS/eCourts, Etc
Benefit-Linked (Non-FIR)	Educational Grant, Housing, Self-Employment, Inter-Caste Marriage, Funeral	Application or Documents Upload via DigiLocker

Adaptive Workflow Engine

The Workflow Engine identifies the **relief type** from the Master Case Object (Layer 1) and initiates the corresponding sub-workflow:

A. Case-Linked Reliefs (3-Stage PoA Model)

Applies to: Immediate Relief, Compensation, Rehabilitation Grant

“Can be modified/updated according to the State’s Disbursements Policy”

Stages:

Stage	Trigger Event	Fund Released	Approval Authority
1	FIR Registered & Verified	25%	DWO (Verification) + DM (Approval)
2	Charge-Sheet Filed	50%	DWO (Validation) + DM (Approval)
3	Conviction / Final Order	25%	DM + Treasury Officer

Flow Summary:

- System pulls case data (FIR, court order, Aadhaar, PFMS) → runs advisory checks
- DWO validates → DM approves → PFMS API triggers payment
- Final disbursement after conviction verified via eCourts
- Every step logged and signed digitally (eSign 2.0 + PFMS confirmation)

Outputs:

- Transparent, human-approved disbursements
 - No automation without oversight
 - Full digital trail across all stages
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B. Benefit-Linked Reliefs (Single or Dual Stage)

Applies to: Educational Assistance, Self-Employment Subsidy, Housing, Inter-Caste Marriage Incentive, Funeral Assistance

“Can be modified/updated according to the State’s Disbursements Policy”

Workflow Adaptation:

Relief Type	Trigger Event	Approval Workflow	Fund Release Logic
Educational Grant	Student uploads verification docs via DigiLocker	DWO verification → DM sanction	100% post-verification
Housing Assistance	Application + Caste/Atrocity Verification	DWO validation → DM approval → Treasury check	60% initial + 40% after housing completion
Self-Employment Subsidy	Business certificate / bank linkage	DWO validation → DM sanction	100% after PFMS account verification
Inter-Caste Marriage	Joint Aadhaar + Marriage Certificate verified	DWO validation → DM sanction	Single-stage disbursement
Funeral Assistance	Death certificate via DigiLocker	DWO validation → DM sanction	Immediate 100% fund transfer

Flow Summary:

- Victim or beneficiary uploads relevant docs (through DigiLocker or manual upload)
- System performs pre-checks → DWO validates
- DM approves final sanction → PFMS triggers direct transfer
- Logs pushed to Layer 4 for audit & to Layer 5 for analytics

Outputs:

- Flexible, workflow-based system supporting all PoA relief categories
 - Human-in-the-loop ensured even for non-FIR schemes
 - Same audit, SLA, and escalation framework used as FIR-linked cases
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End-to-End Execution Flow

Step 1: Case Entry & Relief Selection

1. User logs in (victim or official).
2. Chooses one or more reliefs (Immediate Relief, Housing, etc.).
3. For each selected relief, the system activates a dedicated workflow path.
4. A unified **Case ID** is generated, with linked **Relief IDs** under it.

Step 2: Pre-Validation (System-Assisted, Human-Governed)

- Layer 1 data (FIR, caste, Aadhaar, PFMS) is fetched and verified.
- Hash validation ensures documents are untampered.
- The system only *flags* inconsistencies — it never auto-approves.
- DWO dashboard shows alerts like:
 - Missing document(s)
 - Duplicate claims
 - Mismatched Aadhaar or caste record

Step 3: Human Approval Workflow

Each relief category follows its defined path:

- **For FIR-based Reliefs (3 Stages)**
 - DWO validates → DM approves → Treasury confirms disbursement via PFMS
 - Funds released in 25% / 50% / 25% stages
 - Case moves automatically to next stage on event trigger (FIR → Charge-sheet → Conviction)
- **For Non-FIR Reliefs (1–2 Stages)**
 - User uploads required documents via DigiLocker
 - DWO verifies → DM approves → Treasury disburses (if applicable)
 - All disbursements recorded with PFMS TxIDs

Governance & Design Guarantees

- **Human-controlled, system-assisted** fund release
 - **Dynamic workflow selection** (case vs. benefit)
 - **Digital signatures** for every sanction (eSign 2.0)
 - **Full traceability** in Layer 4 (who verified, who approved, when disbursed)
 - **Automatic SLA tracking** and escalation for delays
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Step 4: Escalations, SLAs & Exception Handling

- **SLA Monitoring:**
 - FIR-based reliefs → Stage 1: 7 days | Stage 2: 14 days | Stage 3: 14 days
 - Non-FIR reliefs → 10–15 days (based on relief type)
 - **Escalation Flow:**
 - Missed SLA → Alert to Collector → 48 hrs delay → Alert to State Secretary
 - All breaches logged to Layer 4 (Audit)
 - **Exception Handling:**
 - Missing docs → Re-verification (Layer 1)
 - FIR mismatch → Manual review
 - Bank/PFMS failure → Retry queue
 - State rule conflict → Sent to Policy Node
 - **Audit:**
 - Every escalation/exception logged with timestamp, officer ID & action trace.
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Step 5: Notifications & Citizen Updates

- Every stage approval and fund transfer triggers:
 - SMS via **NIC SMS Gateway**
 - Email via **DigiMail API**
 - Live status update on the **Citizen Dashboard** (Stage 1, Stage 2, Stage 3 with timestamps and Transaction IDs).
 - Citizens can also download **Stage-wise Sanction Certificates** (auto-generated PDFs signed by officials).
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Step 6: Audit, Logging & Data Handoff

1. Every workflow action (approval, disbursement, rejection) generates a log entry (timestamp, officer ID, digital hash).
 2. Logs are pushed to **Layer 4 – Audit Layer** for transparency.
 3. Layer 2 also passes summary reports to **Layer 5 – Analytics Layer** for dashboard metrics and fund utilization analysis.
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Core Subsystems — Layer 2

Component	Function
Workflow Engine	Manages case flow for all user-selected relief types.
Rules & SLA Engine	Enforces eligibility checks and timelines per relief stage.
Human Approval Gateway	Interface for officials to review and approve cases.
Notification Service	Sends updates to citizens and officers.
Escalation Module	Alerts higher authorities when SLAs are missed.
PFMS Connector	Triggers verified fund disbursements.
Exception Handler	Handles missing docs or validation failures.

Key Principles Maintained

- **No fund release without human approval.**
 - **Full auditability** of every step (digitally signed + timestamped).
 - **Role-based access** (DM, DWO, Treasury have separate workflows).
 - **Automated progression** but **human-governed final sanction.**
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LAYER 3 — Financial Disbursement & Fund Tracking Layer

“Stage-wise DBT integration uses high-availability PFMS connectors with automatic retry and fallback queues.”

Role:

Securely disburses funds for **user-selected reliefs**, tracks every transaction, and provides real-time visibility. Handles variable stage disbursements depending on relief type and state-specific rules, with officer verification logged for each stage.

□ Core Components

1. PFMS Integration Module

- Connects with PFMS APIs for direct benefit transfer.
- Validates Aadhaar-linked beneficiary accounts.
- Executes payments via Aadhaar Payment Bridge (APB).
- Stage-specific triggers based on relief type:
 - Stage 1 → Initial approval by DM/DWO (if applicable).
 - Stage 2 → Mid-stage approval by DWO/DM (if applicable).
 - Stage 3 → Final approval by DM + Treasury Officer (if applicable).

2. Transaction Tracker

- Monitors all payment statuses: Success, Pending, Failed.
- Records stage-wise transaction IDs with officer approvals.
- Maps transactions to relief type selected by the citizen.

3. Retry & Fallback Queue

- Automatically retries failed transactions.
- Logs retries with timestamps for audit and grievance handling.
- Ensures funds are not lost due to transient failures.

4. Treasury Ledger Sync

- Updates district/state treasury accounts with stage-wise fund utilization.
- Tracks remaining fund availability for subsequent stages.
- Provides cumulative visibility for multiple relief types.

Workflow

1. Layer 2 triggers PFMS API call after human approvals per stage.
 2. Beneficiary Aadhaar-linked account validation completed.
 3. Funds disbursed → confirmation logged in Layer 4 audit system.
 4. Stage-wise status updated in citizen dashboard and internal case management.
 5. Multiple reliefs selected by user are tracked separately, ensuring each relief's disbursement aligns with its applicable stage(s).
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LAYER 4 — Audit, Transparency & Grievance Layer

“Logs and grievance management are replicated and fault-tolerant, ensuring transparency and SLA tracking.”

Role:

Maintains full traceability and accountability for all actions, including pre-checks, verifications, and disbursement. Manages citizen grievances and ensures transparent reporting.

Core Components:

1. Event Logging System

- Logs every action: DWO verification, DM approvals, Treasury checks, PFMS disbursement.
- Stage-wise tagging (Stage 1/2/3).
- Each log digitally signed and stored in WORM-compliant storage.
- **Implements time-bound data retention (as per DPDP Act, 2023). Case files and KYC data are auto-purged or archived after fund closure.**

2. Audit Dashboard

- Web-based view for auditors and monitoring cells.
- Filters by stage, officer, district, and case.
- Highlights delays, pending approvals, and failed transactions.

3. Grievance Engine

- Citizens can submit complaints via web or helpline.
- Tickets auto-assigned to relevant officers (DWO or DM).
- Escalation based on delay per stage.

4. Integrity Checker

- Periodic hash verification to detect tampering of case files, approval notes, and fund logs.

LAYER 5 — Analytics & Insights Layer

Role:

Provides operational intelligence, monitors stage-wise and relief-specific fund disbursement efficiency, evaluates officer performance, and ensures transparency for governance and citizens.

Core Components

- 1. Data Warehouse (PostgreSQL + ClickHouse)**
 - Stores complete case history, user-selected relief types, stage-wise disbursement records, officer approvals, and audit logs.
 - Maintains mapping between relief types, stages, and state-specific rules.
- 2. Analytics Engine (Python / PowerBI / Metabase)**
 - Generates metrics for governance and operational monitoring:
 - Average time per stage and per relief type (Stage 1, 2, 3 or 1–2 as per state rules)
 - Pending cases by stage, district, and relief type
 - Fund utilization rate by stage and relief type
 - Officer efficiency: DWO verification vs. DM/Treasury approval timelines
 - SLA adherence and escalation trends
 - Highlights bottlenecks and inefficiencies for administrative intervention.
- 3. Predictive Module (Optional AI)**
 - Suggests districts or relief types requiring proactive monitoring.
 - Forecasts fund requirements and potential case surges.
 - Officer-driven decisions remain final; system only provides recommendations.
- 4. Open Data API**
 - Provides anonymized, stage-wise, and relief-specific statistics for public dashboards, research, and policy planning.
 - Ensures citizen transparency while preserving sensitive personal data.

Cross-Layer Security

- End-to-end encryption (TLS 1.3).
- AES-256 at rest.
- Role-based access control: DWO, DM, Treasury Officer, Citizen.
- DPDP & GIGW compliance.
- Regular penetration and vulnerability testing.

- **Implements centralized KMS (Key Management Service) for key rotation and data encryption governance.**

